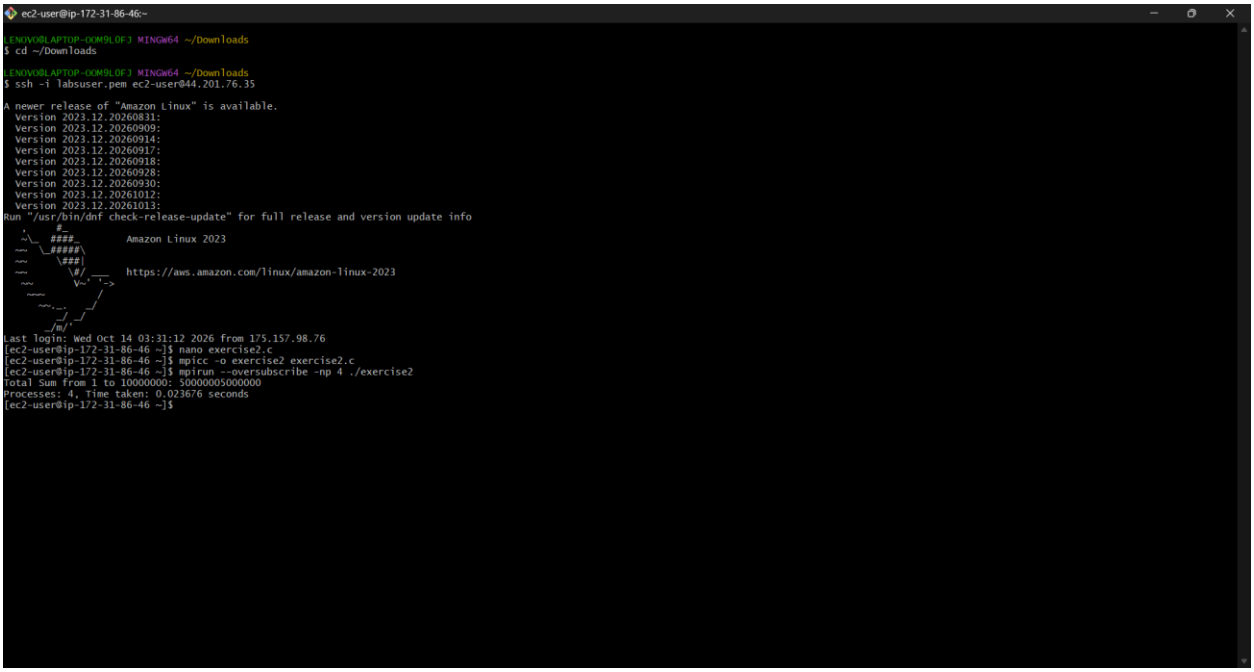
SE3082 - Parallel Computing

Lab 06

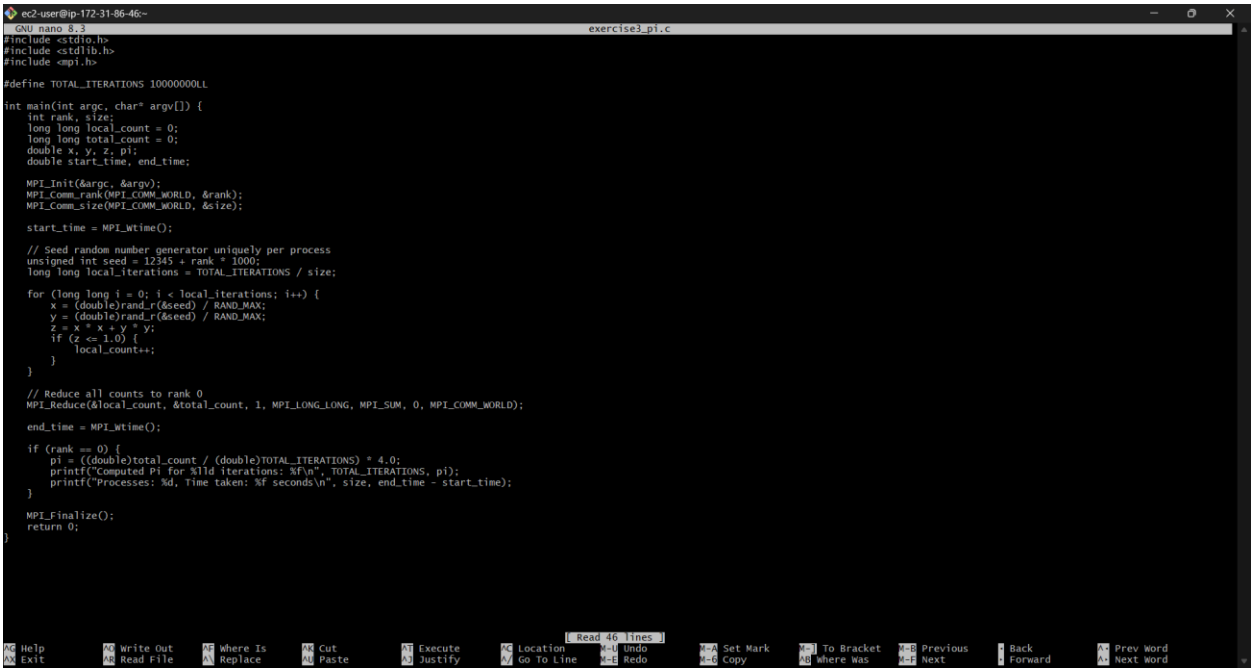
IT24610782 – M.A.V.E. MUNASINGHE

Exercise 2: Parallel Sum





Exercise 3: Monte Carlo Pi Calculation



```
ec2-user@ip-172-31-86-46:~
 GNU nano 8.3                                    exercise3_pi.c
#include <stdio.h>
#include <stdlib.h>
#include <mpi.h>

#define TOTAL_ITERATIONS 10000000LL

int main(int argc, char* argv[]) {
    int rank, size;
    long long local_count = 0;
    long long total_count = 0;
    double x, y, z, pi;
    double start_time, end_time;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    start_time = MPI_Wtime();

    // Seed random number generator uniquely per process
    unsigned int seed = 12345 + rank * 1000;
    long long local_iterations = TOTAL_ITERATIONS / size;

    for (long long i = 0; i < local_iterations; i++) {
        x = (double)rand_r(&seed) / RAND_MAX;
        y = (double)rand_r(&seed) / RAND_MAX;
        z = x * x + y * y;
        if (z <= 1.0) {
            local_count++;
        }
    }

    // Reduce all counts to rank 0
    MPI_Reduce(&local_count, &total_count, 1, MPI_LONG_LONG, MPI_SUM, 0, MPI_COMM_WORLD);

    end_time = MPI_Wtime();

    if (rank == 0) {
        pi = ((double)total_count / (double)TOTAL_ITERATIONS) * 4.0;
        printf("Computed Pi for %lld iterations: %f\n", TOTAL_ITERATIONS, pi);
        printf("Processes: %d, Time taken: %f seconds\n", size, end_time - start_time);
    }

    MPI_Finalize();
    return 0;
}

                                [ Read 46 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark    M-] To Bracket  M-B Previous  ^← Back         ^Space Prev Word
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy        ^Q Where Was    M-F Next      ^→ Forward      ^Z Next Word
```



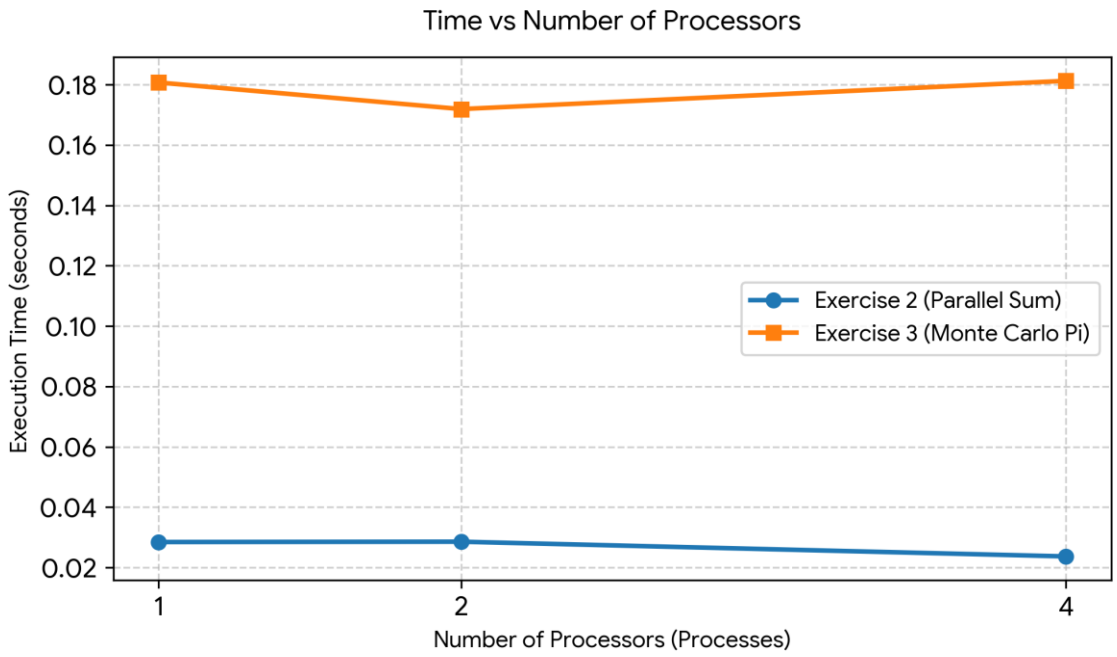
```
ec2-user@ip-172-31-86-46:~
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise2

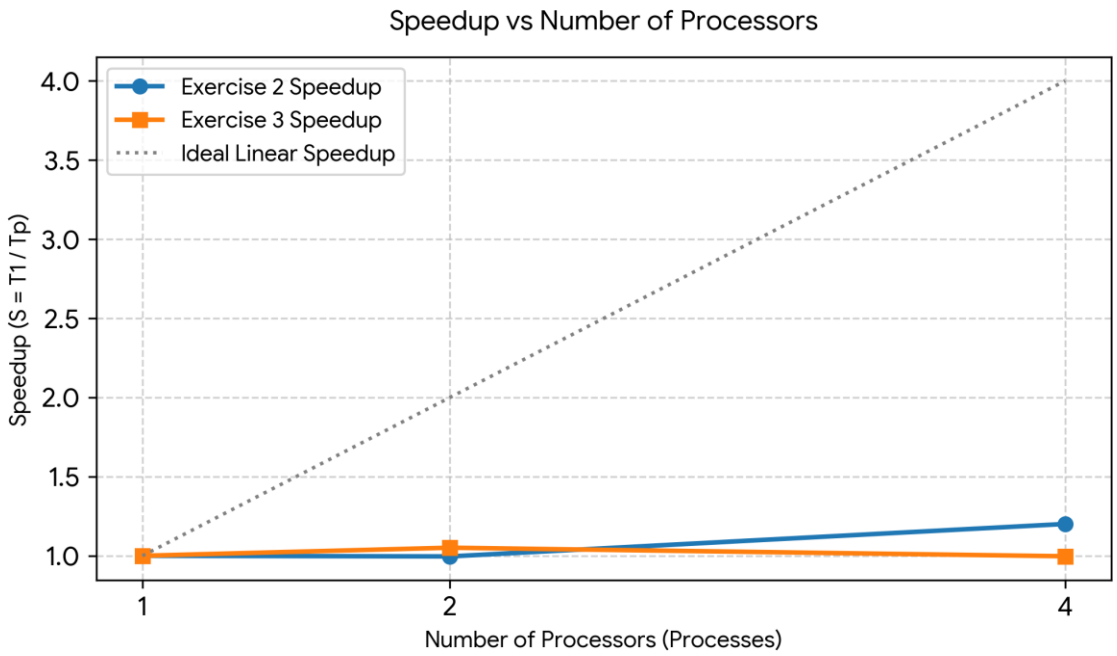
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise2
Total Sum from 1 to 10000000: 50000005000000
Processes: 2, Time taken: 0.028555 seconds
[ec2-user@ip-172-31-86-46 ~]$ nano exercise3_pi.c
[ec2-user@ip-172-31-86-46 ~]$ mpicc -o exercise3_pi exercise3_pi.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 1 ./exercise3_pi
Computed Pi for 10000000 iterations: 3.141386
Processes: 1, Time taken: 0.180719 seconds
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise3_pi
Computed Pi for 10000000 iterations: 3.140880
Processes: 2, Time taken: 0.171914 seconds
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 4 ./exercise3_pi
Computed Pi for 10000000 iterations: 3.141423
Processes: 4, Time taken: 0.181271 seconds
[ec2-user@ip-172-31-86-46 ~]$
```

Exercise 4: Performance Evaluation & Speedup Analysis

```
ec2-user@ip-172-31-86-46:~
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Oct 14 03:31:12 2026 from 175.157.98.76
[ec2-user@ip-172-31-86-46 ~]$ nano exercise2.c
[ec2-user@ip-172-31-86-46 ~]$ mpicc -o exercise2 exercise2.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 4 ./exercise2
Total Sum from 1 to 10000000: 50000005000000
Processes: 4, Time taken: 0.023676 seconds
[ec2-user@ip-172-31-86-46 ~]$ nano exercise2.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 1 ./exercise2
Total Sum from 1 to 10000000: 50000005000000
Processes: 1, Time taken: 0.028443 seconds
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise2

[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise2
Total Sum from 1 to 10000000: 50000005000000
Processes: 2, Time taken: 0.028555 seconds
[ec2-user@ip-172-31-86-46 ~]$
```





Exercise 5: Communication Mismatch & Buffered Send

Part 1: Source and Destination Mismatch



```
ec2-user@ip-172-31-86-46:~
 GNU nano 8.3                                  exercise5_mismatch.cc
#include <mpi.h>
#include <iostream>

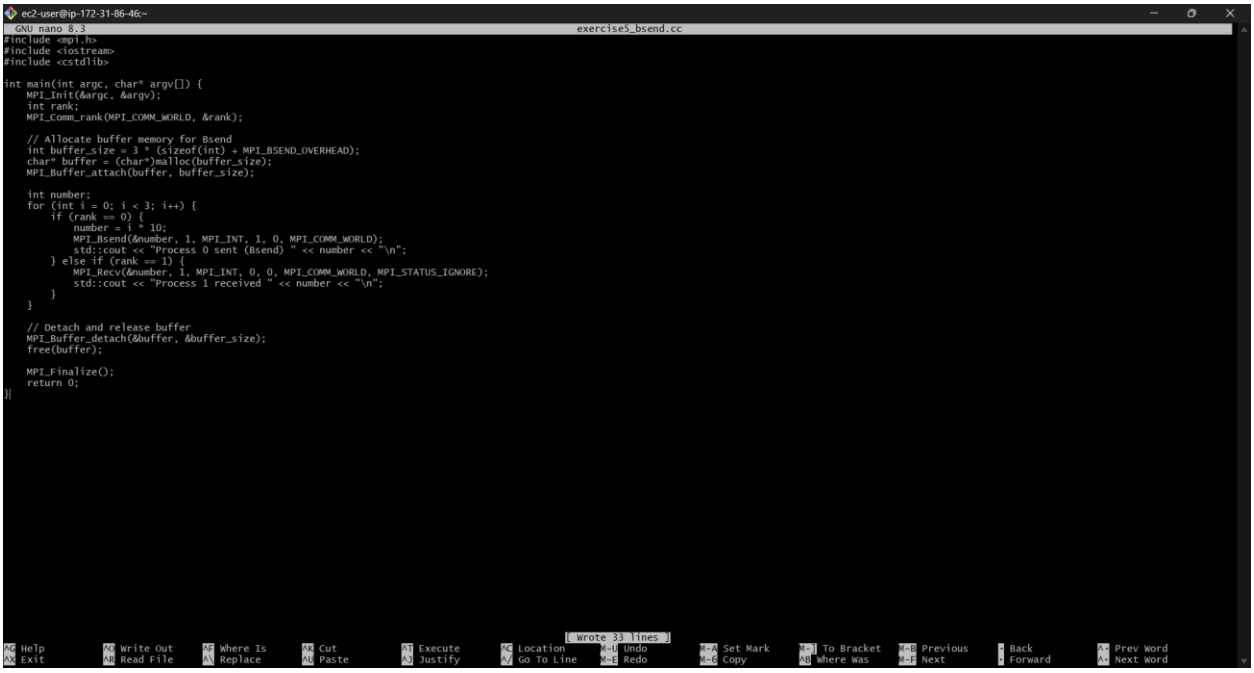
int main(int argc, char* argv[]) {
    MPI_Init(&argc, &argv);
    int rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);

    int number;
    if (rank == 0) {
        number = 42;
        // Rank 0 sends to Rank 1
        MPI_Send(&number, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
        std::cout << "Process 0 sent " << number << "\n";
    } else if (rank == 1) {
        // Rank 1 expects to receive from Rank 2 (which does not send anything!)
        MPI_Recv(&number, 1, MPI_INT, 2, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
        std::cout << "Process 1 received " << number << "\n";
    }

    MPI_Finalize();
    return 0;
}
                                [ Read 23 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark    M-] To Bracket  M-B Previous  ^← Back        ^A Prev Word
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy        ^Q Where Was    M-F Next      ^→ Forward     ^▲ Next Word
```

```
Complete!
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_mismatch exercise5_mismatch.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 3 ./exercise5_mismatch
Process 0 sent 42
```

Part 2: Implementation of Buffered Send




```
nano exercise5_bsend.cc
^C[ec2-user@ip-172-31-86-46 ~]$ nano exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_bsend exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise5_bsend
Process 0 sent (Bsend) 0
Process 0 sent (Bsend) 10
Process 0 sent (Bsend) 20
Process 1 received 0
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-86-46 ~]$
```

Exercise 6: Receiving with MPI_ANY_SOURCE



```
ec2-user@ip-172-31-86-46:~
 GNU nano 8.3                                    exercise6_anysource.c
#include <stdio.h>
#include <stdlib.h>
#include <mpi.h>

#define TOTAL_ITERATIONS 10000000LL

int main(int argc, char* argv[]) {
    int rank, size;
    long long local_count = 0;
    long long total_count = 0;
    double x, y, z, pi;
    double start_time, end_time;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    start_time = MPI_Wtime();

    unsigned int seed = 12345 + rank * 1000;
    long long local_iterations = TOTAL_ITERATIONS / size;

    for (long long i = 0; i < local_iterations; i++) {
        x = (double)rand_r(&seed) / RAND_MAX;
        y = (double)rand_r(&seed) / RAND_MAX;
        z = x * x + y * y;
        if (z <= 1.0) {
            local_count++;
        }
    }

    if (rank != 0) {
        // Send partial count to rank 0
        MPI_Send(&local_count, 1, MPI_LONG_LONG, 0, 0, MPI_COMM_WORLD);
    } else {
        total_count = local_count;
        long long received_count = 0;
        MPI_Status status;

        // Receive from any worker process using MPI_ANY_SOURCE
        for (int i = 1; i < size; i++) {
            MPI_Recv(&received_count, 1, MPI_LONG_LONG, MPI_ANY_SOURCE, 0, MPI_COMM_WORLD, &status);
            total_count += received_count;
            printf("Rank 0 received %lld points from Rank %d\n", received_count, status.MPI_SOURCE);
        }
    }

    end_time = MPI_Wtime();

    if (rank == 0) {
        pi = ((double)total_count / (double)TOTAL_ITERATIONS) * 4.0;
        printf("Computed Pi (MPI_ANY_SOURCE) for %lld iterations: %f\n", TOTAL_ITERATIONS, pi);
        printf("Processes: %d, Time taken: %f seconds\n", size, end_time - start_time);
    }

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark   M-] To Bracket  M-B Previous   ^← Back        ^↑ Prev Word
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy       ^B Where Was    M-F Next       ^→ Forward     ^↓ Next Word
```

```
Complete!
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_mismatch exercise5_mismatch.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 3 ./exercise5_mismatch
Process 0 sent 42

nano exercise5_bsend.cc
^C[ec2-user@ip-172-31-86-46 ~]$ nano exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_bsend exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise5_bsend
Process 0 sent (Bsend) 0
Process 0 sent (Bsend) 10
Process 0 sent (Bsend) 20
Process 1 received 0
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-86-46 ~]$ nano exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ nano exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ mpicc -o exercise6_anysource exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 4 ./exercise6_anysource
Rank 0 received 1963446 points from Rank 1
Rank 0 received 1964298 points from Rank 3
Rank 0 received 1963353 points from Rank 2
Computed Pi (MPI_ANY_SOURCE) for 10000000 iterations: 3.141423
Processes: 4, Time taken: 0.178824 seconds
[ec2-user@ip-172-31-86-46 ~]$
```

Exercise 7: Buffered Send (MPI_Bsend) with MPI_ANY_SOURCE




```
Complete!
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_mismatch exercise5_mismatch.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 3 ./exercise5_mismatch
Process 0 sent 42

nano exercise5_bsend.cc
^C[ec2-user@ip-172-31-86-46 ~]$ nano exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpicxx -o exercise5_bsend exercise5_bsend.cc
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 2 ./exercise5_bsend
Process 0 sent (Bsend) 0
Process 0 sent (Bsend) 10
Process 0 sent (Bsend) 20
Process 1 received 0
Process 1 received 10
Process 1 received 20
[ec2-user@ip-172-31-86-46 ~]$ nano exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ nano exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ mpicc -o exercise6_anysource exercise6_anysource.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 4 ./exercise6_anysource
Rank 0 received 1963446 points from Rank 1
Rank 0 received 1964298 points from Rank 3
Rank 0 received 1963353 points from Rank 2
Computed Pi (MPI_ANY_SOURCE) for 10000000 iterations: 3.141423
Processes: 4, Time taken: 0.178824 seconds
[ec2-user@ip-172-31-86-46 ~]$ nano exercise7_bsend.c
[ec2-user@ip-172-31-86-46 ~]$ mpicc -o exercise7_bsend exercise7_bsend.c
[ec2-user@ip-172-31-86-46 ~]$ mpirun --oversubscribe -np 4 ./exercise7_bsend
Rank 0 received (via Bsend) 1964298 points from Rank 3
Rank 0 received (via Bsend) 1963353 points from Rank 2
Rank 0 received (via Bsend) 1963446 points from Rank 1
Computed Pi (Exercise 7 - Bsend) for 10000000 iterations: 3.141423
Processes: 4, Time taken: 0.165406 seconds
[ec2-user@ip-172-31-86-46 ~]$
```